Summary in Graph

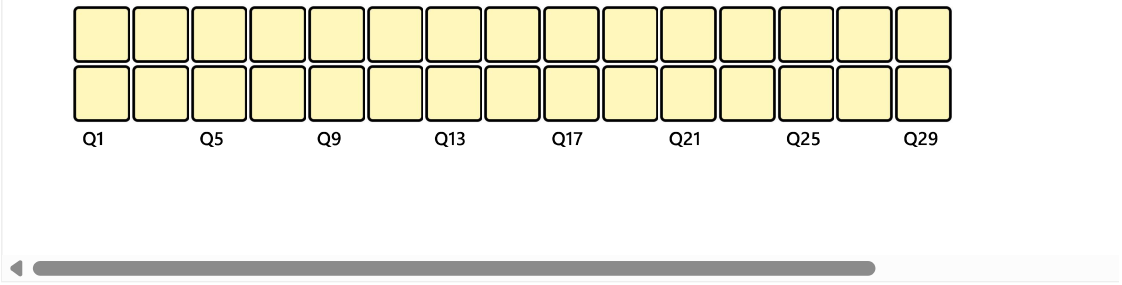
Exam Summary (GO Classes DA Test Series 2026 | Data Structures & Algorithms | Subject Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	81 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (17) out of 174 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python program.

```
    def func(n):
        if n == 0:
            print("0")
        elif n == 1:
5.          print("1")
        else:
            func(n//2)
            print(n % 2)
```

What will be the output of `func(128)`?

A. 0

B. 1

C. 10000000

D. 100

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose you have a stack in which the values 1 through 5 must be pushed on the stack in that order, but that an item on the stack can be popped at any time. Which of the following is valid sequence of push and pop operations such that the values are popped in the following order:

i. 23541
ii. 43521
iii. 21534

A. (i) and (ii)
B. (ii) and (iii)
C. (i) and (iii)
D. (i), (ii) and (iii)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following algorithm to check whether a string of parentheses is valid and balanced:

- Whenever an opening bracket ' (' is encountered, it is pushed onto the stack.
- Whenever a closing bracket ') ' is encountered, a bracket is popped from the stack.

What is the maximum number of characters that will appear on the stack at any time when the algorithm analyzes the string: (()()((()))(()))

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following pseudocode of a recursive function that takes a pointer to Node which is the head of a NULL-terminated singly linked list, and a key x.

```
Procedure(head, key):
    If (head == NULL) return NULL

    If (head.key == x)
        node *temp = head.next
        delete head
        return Procedure(temp, key)

    Else
        head.next = Procedure(head.next, key)
        return head;
```

Which of the following is True about the function?

A. The function deletes the first node of a singly linked list that matches the given key.
B. The function deletes every node of a singly linked list that matches the given key.
C. The function attempts to delete the first node of a singly linked list that matches but the implementation is wrong.
D. The function attempts to delete every node of a singly linked list that matches but the implementation is wrong.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

The following nonrecursive partial pseudocode reverses a list represented with a doubly linked list using a single pass through the list.

head: points to the head of doubly linked list.
prev: node's pointer to the previous node.
next: node's pointer to the next node.




```
Procedure(head):
    if (head == NULL or head.next == NULL) return

    previousNode = head
    currentNode = previousNode.next

    While ( _____(i)_____ ) {
        nextNode = currentNode.next

        currentNode.prev = _______(ii)_______
        currentNode.next = previousNode

        previousNode = currentNode
        currentNode = nextNode
     }
```

Which of the following are appropriate choice for (i) and (ii) respectively?

A. (i) `currentNode != NULL`
B. (ii) `currentNode.next`
C. (i) `currentNode.next != NULL`
D. (ii) `nextNode`

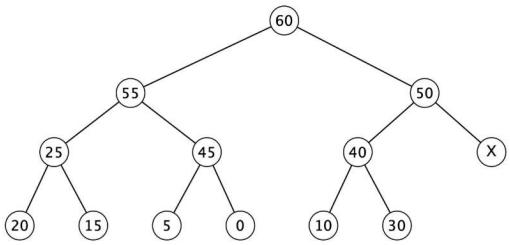
Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 06sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Suppose we have 4 distinct positive keys. All possible binary heaps are constructed by using these keys. let 'A' denotes the number of max-heap and ' B ' denotes the number of min-heap constructed, then $A+3B=$ _____

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures



If the last inserted key in Heap is 50 then what is the possible value(s) of X?

A. 35
B. 40
C. 45
D. 50

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

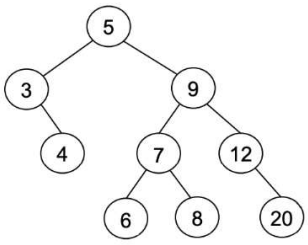
Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

The number of ways to create a binary heap tree with elements $21, 23, 53, 22, 14, 15, 90, 32$?

Your Answer: | Correct Answer: 210 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The binary search tree shown below was constructed by inserting a sequence of items into an empty tree



Which of the following input sequences will not produce this binary search tree?

A. 5
3
4
9
12
7
8
6
20

B. 5
9
3
7
6
8
4
12
20

C. 5
9
7
8
6
12
20
3
4

D. 5
9
3
6
7
8
4
12
20

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a binary search tree (BST) with n elements and height $h = O(\log(n))$, the correct asymptotic bound to find the $k\text{-}th (k < n)$ largest element in the tree is _____

A. $O(\mathrm{k})$
B. $O(\mathrm{h})$
C. $O(\mathrm{h}+\mathrm{k})$
D. $O(\mathrm{n})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider a Binary Search Tree (BST). Which of the following statements is/are TRUE regarding the inorder predecessor and inorder successor of a node being deleted?

A. Inorder predecessor of a deleting node is always either a leaf node or node without the right child.
B. Inorder successor of a deleting node is always either a leaf node or node without the right child.
C. Inorder predecessor of a deleting node is always either a leaf node or node without the left child.
D. Inorder successor of a deleting node is always either a leaf node or node without the left child.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A rooted tree is called a 3-ary tree if every node has either 0 or 3 children. Consider such a 3 -ary tree with n total nodes. If the number of leaf nodes in the tree is L, then the number of internal nodes in the tree is:

A. $\frac{L-1}{2}$

B. $\frac{L-1}{3}$

C. $\frac{3L-1}{2}$

D. $\frac{n-1}{L}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

A System of difference constraints has inequalities of the form $\mathrm{x}_j - \mathrm{x}_i <= \mathrm{C}$. These can be represented using a directed graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$, where each variable is a node and each constraint becomes an edge x_j to x_i with weight C.

Consider the following system of constraints:

$x_1 - x_2 <= 2$

$x_2 - x_3 <= 1$

$x_3 - x_1 <= -3$

$x_1 - x_4 <= 4$

What can you conclude about the system?

A. The system has no feasible solution because of the negative cycle.

B. The system has unique feasible solution

C. The system has infinitely many feasible solutions.

D. The system has feasible solutions only if $x_1 = x_2 = x_3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following statements about the Depth-first search is/are TRUE?

- **S1**. The running time of depth-first search in a complete graph, as a function of $|V|$ and $|E|$, if the input graph is represented by an adjacency list, is $O(|V|^2)$

- **S2**. If the depth-first search tree of an undirected graph $G(V, E)$ has exactly one back edge and no other edges then the G has exactly one cycle.

A. Both S1 and S2 are True

B. S1 is True and S2 is False

C. S1 is False and S2 is True

D. Both S1 and S2 are False

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the recurrence relation:

$$T(n) = T(n-1) + \Theta(\log n);$$

$$T(1) = \theta(1)$$

Determine the asymptotic time complexity of $T(n)$.

A. $\Theta(\log n)$

B. $\Theta(n)$

C. $\Theta(n \log n)$

D. $\Theta(n^2)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider an array based implementation of a stack, and suppose that it is initially empty. Upon n push operations the array will be resized in such a way that the running time per push is $O(1)$ per operation, on average. How many times is the array resized over the n pushes, using this scheme?

A. $O(1)$

B. $O(\log n)$

C. $O(\mathrm{n})$

D. $O\left(n^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following statements for the simple undirected graph $G(V, E)$:

- **S1**: For any vertex $v \in G$ if v has more than child in depth first tree(DFT) then v is an articulation point in G.
- **S2**: For any vertices $\mathrm{u}, \mathrm{v}, \mathrm{w} \in \mathrm{G}$ if v is ancestor of u and w is descendent of u in DFT(depth Fiirst Tree), then there exists a cycle in G.
- **S3**: for any vertex u in DFT if v is ancestor of u and there exists no back edge from u or its any descendent then edge $u - v$ is a bridge.

Which of the following is/are True?

A. only S1 is True
B. only S3 is True
C. Both S1 and S3 are True
D. All S1, S2, S3 are True

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

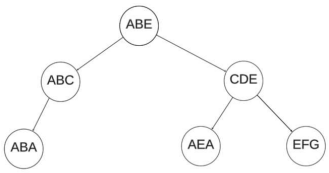
For a simple connected undirected graph $G = (V, E)$, which of the following statements are True for dijastra's algorithm?

A. Each vertex in the algorithm is relaxed for at least one time.
B. If the length of the shortest path from source vertex s to v is less than the shortest path s to w then vertex v is relaxed before vertex w.
C. During each edge relaxation, for each vertex v , distTo $[\mathrm{v}]$ decreases.
D. Immediately after relaxing any edge $v \rightarrow w$, distTo[w] is the length of the shortest path from s to w .

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a Blanched Binary Search Tree of **n** students' with keys as their names, stored in lexicographical ordering. Let k be the number of students whose name starts with prefix **X**.
For eg. a class having 6 students with names { ABE , CDE , AEA , ABC , EFG , ABA}, there are $\mathrm{k} = 3$ students whose name starts with prefix X=AB.



Which of the following is the correct time complexity to find all the students with prefix X.

A. $\theta(n \cdot \log \mathrm{k} + \mathrm{k})$
B. $\theta(n + \mathrm{k})$
C. $\theta(\log n + \mathrm{k})$
D. $\theta(\mathrm{k} \cdot \log \mathrm{n} + \mathrm{k})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a hash table with n buckets, where linear probing is used to resolve collisions. The hash function is such that the probability that a key value is hashed to a particular bucket is $\frac{1}{n}$. The hash table is initially empty and distinct values are inserted in the table.
What is the probability that the given two distinct keys will be inserted at consecutive slots successfully?

A. $\frac{2}{n^2}$

B. $\frac{2}{n}$

C. $\frac{3}{n}$

D. $\frac{3}{n^2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a hash function that distributes keys uniformly. The hash table contains m slots, if chaining is used for collision resolution then the total number of available probe sequences for n keys?

A. $\mathrm{m!}$

B. m^2

C. m^n

D. m

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the recursive quicksort algorithm with "random pivoting". That is, in each recursive call, a pivot is chosen uniformly at random from the sub-array being sorted. When this randomized algorithm is applied to an array of size "n" all whose elements are distinct, what is the probability that the smallest and the largest elements in the array are compared during a run of the algorithm?

A. $\frac{1}{n}$

B. $\frac{2}{n}$

C. $\theta(\frac{1}{n.logn})$

D. $\theta(\frac{1}{n^2})$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements are TRUE.

A. While sorting an array A of n elements using merge sort, the number of element comparisons at each level of the merge tree is $\Theta(\log n)$ in the worst case.

B. In worst-case quicksort, the number of comparisons is equal to $\Theta\left(n^2\right)$, Which cause the recursion tree to be grow to a depth of $n-1$

C. The average number of swaps in selection sort is $\Theta(n)$ regardless of input size.

D. Heapsort can be implemented to be stable and in-place simultaneously.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

An array A of n real distinct numbers is said to be quadratically true if the elements of the array represent values of some quadratic function, i.e.,

$$A[i] = ai^2 + bi + c \quad \text{for constants } a, b, c \text{ and } 0 \leq i < n$$

Assume that $\mathrm{a}, \mathrm{c} > 0$ and $\mathrm{b} < 0$, the function represents a convex parabola, and the array has a single local minimum.
What is the best possible asymptotic bound to find the index of the local minimum in such an array?

A. $\theta(logn)$

B. $\theta(n)$

C. $\theta(1)$

D. $\theta\left(\log^2 \mathrm{n}\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $A = [a0, a1, a2, \ldots, an]$ be an unsorted array where n is a power of 2. Suppose Quick Sort is applied to this array, and at every recursive step, the partition function always divides the array in 1:n. What will be the time complexity of the Quick Sort implementation?

A. $\Theta(\mathrm{n})$

B. $\Theta(nlogn)$

C. $\Theta\left(n^2\right)$

D. $\Theta(\log n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE?

A. A DFS tree always has exactly $\mathrm{n}-1$ edges if the graph G has n vertices.

B. Let T be a depth -first search tree of an undirected graph G . Let (x, y) be an edge of G that is not an edge of T, then one of x or y is an ancestor of the other.

C. If an edge (u, v) in an undirected graph G is not a part of the DFS tree and neither u is an ancestor of v nor v is an ancestor of u in the DFS tree, then the edge (u, v) is a cross edge.

D. In a DFS if vertex u is discovered before vertex v then there is an edge (u, v) in G and either u or v are ancestors of others in the DFS tree.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider procedures 1 and 2 expressed in pseudocode as Follows:

```
begin procedure 1
    while n > 1 do
        for i = 1 to n  do
            printf("hello")
        end for
        n = floor(n/2)
    end while
end procedure
```

```
begin procedure 2
    begin If n > 1
        printf("hello")
        begin
            procedure 2(floor(n/2))
        end
    end If
end procedure
```

Let $\mathrm{f1(n)}$ and $\mathrm{f2(n)}$ denote the number of times the statement "printf("hello")" executed in procedure 1 and procedure 2, respectively.
Which of the following statements is/are TRUE?

A. $f1 \in O(f2)$

B. $f1 \in \Omega(f2)$

C. $f2 \in O(f1)$

D. $f1 \in \theta(f2)$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{3}\right) + T\left(\frac{n}{6}\right) + O(n)$$
$$T(n) = 5 \quad \forall n < 100$$

Which of the following is the correct aysmptotic growth of $T(n)$?

A. $T(n) = \Theta(n \log n)$

B. $T(n) = \Theta\left(n \log^2 n\right)$

C. $T(n) = \Theta\left(n^2\right)$

D. $T(n) = \Theta\left(n^2 \log n\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Which of the following recurrence relation claims is/are TRUE:

I. $\mathrm{T}(n) = 3\mathrm{T}(n/6) + cn$, for some positive constant c, and $\mathrm{T}(1) = 1$, then $\mathrm{T}(n) = O(n)$.
II. $\mathrm{T}(n) = \mathrm{T}(n/2) + \log n$, and $\mathrm{T}(1) = 1$, then $\mathrm{T}(n) = O(\log n)$.

A. Both (I) and (II) is True
B. (I) is True and (II) is False
C. (I) is False and (II) is True
D. Both (I) and (II) is False

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose that A is an array of n elements. Each element of A has some flavour; you cannot tell the flavours apart, but you know that one flavour is in the strict majority. A flavour is said to be in strict majority if more than n/2 elements are the same. You have access to a genie called isMajority so that isMajority returns TRUE if x is an element of A with majority flavour and FALSE otherwise.

```
def findMajority(A):
  n = len(A) # assume that n is a power of 2.
  if n == 1:
    return A[0]
  a = findMajority(A[:n//2])
  b = findMajority(A[n//2:])
  if isMajority(A, a):
    return a
  else:
    return b
```

Suppose that $isMajority$ runs in time $\theta(\sqrt{n})$ on an array of length n. What is the running time of $findMajority$?

A. $\theta(\sqrt{n})$

B. $\theta(n)$

C. $\theta(n.logn)$

D. $\theta(n2)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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